Solutions to Axler's Linear Algebra

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Contents

3D Invertibility and Isomorphisms

Exercise 3D1

Suppose $T \in \mathcal{L}(V, W)$ is invertible. Show that T^{-1} is invertible and

$$(T^{-1})^{-1} = T.$$

Solution. Since T is invertible, then $T^{-1} \in \mathcal{L}(W, V)$. By $TT^{-1} = I$ and $T^{-1}T = I$, we have that T^{-1} is invertible and $(T^{-1})^{-1} = T$. ■

Exercise 3D2

Suppose $T \in \mathcal{L}(U, V)$ and $S \in \mathcal{L}(V, W)$ are both invertible linear maps. Prove that $ST \in \mathcal{L}(U, W)$ is invertible and that $(ST)^{-1} = T^{-1}S^{-1}$.

Solution. Since T and S are invertible, then $T^{-1} \in \mathcal{L}(V, U)$ and $S^{-1} \in \mathcal{L}(W, V)$. By the definition of the inverse, we have that

$$(ST)(T^{-1}S^{-1}) = S(TT^{-1})S^{-1} = SIS^{-1} = SS^{-1} = I,$$

and

$$(T^{-1}S^{-1})(ST) = T^{-1}(S^{-1}S)T = T^{-1}IT = T^{-1}T = I.$$

Thus, ST is invertible and $(ST)^{-1} = T^{-1}S^{-1}$. ■

Exercise 3D3

Suppose V is finite-dimensional and $T \in \mathcal{L}(V)$. Prove that the following are equivalent.

(a) T is invertible.

(b) $Tv_1, \ldots, Tv_n$ is a basis of V for every basis $v_1, \ldots, v_n$ of V.

(c) $Tv_1, \ldots, Tv_n$ is a basis of V for some basis $v_1, \ldots, v_n$ of V.

Solution.

- (a) $\Longrightarrow$ (b): Suppose T is invertible. Then $T^{-1} \in \mathcal{L}(V)$. Let $v_1, \ldots, v_n$ be a basis of V. Since T^{-1} is invertible, then $T^{-1}v_1, \ldots, T^{-1}v_n$ is a basis of V. Then there exist scalars $\alpha_1, \ldots, \alpha_n$ such that

$$v = \alpha_1 T^{-1}v_1 + \cdots + \alpha_n T^{-1}v_n$$

for every $v \in V$. Applying T to both sides, we have

$$Tv = \alpha_1 v_1 + \cdots + \alpha_n v_n$$

for every $v \in V$. Thus, $Tv_1, \ldots, Tv_n$ spans V. Now suppose that

$$\alpha_1 Tv_1 + \cdots + \alpha_n Tv_n = 0$$

for some scalars $\alpha_1, \ldots, \alpha_n$. Applying T^{-1} to both sides, we have

$$\alpha_1 v_1 + \cdots + \alpha_n v_n = 0.$$

Since $v_1, \ldots, v_n$ is a basis of V, then $\alpha_1 = \cdots = \alpha_n = 0$. Hence, $Tv_1, \ldots, Tv_n$ is linearly independent. Since $Tv_1, \ldots, Tv_n$ spans V and is linearly independent, then $Tv_1, \ldots, Tv_n$ is a basis of V.

- (b) $\implies$ (c): This is trivial.
- (c) $\implies$ (a): Suppose $Tv_1, \ldots, Tv_n$ is a basis of V for some basis $v_1, \ldots, v_n$ of V. Then $\operatorname{range} T = V$. By the rank-nullity theorem, we have that $\dim \operatorname{null} T = 0$. Thus, T is injective. Since T is injective and $\operatorname{range} T = V$, then T is invertible. ∎

Exercise 3D4

Suppose V is finite-dimensional and $\dim V > 1$. Prove that the set of non-invertible linear maps from V to itself is not a subspace of $\mathcal{L}(V)$.

Solution. Let $v_1, \ldots, v_n$ be a basis of V. Define $T, S \in \mathcal{L}(V)$ such that

$$Tv_1 = 0, \quad Tv_k = v_k \text{ for } k = 2, \ldots, n,$$

and

$$Sv_1 = v_1, \quad Sv_k = 0 \text{ for } k = 2, \ldots, n.$$

Then T and S are non-invertible linear maps, since $\operatorname{null} T = \operatorname{span} v_1$ and $\operatorname{null} S = \operatorname{span}(v_2, \ldots, v_n)$. However, $T + S$ is invertible, since

$$(T + S)v_1 = v_1, \quad (T + S)v_k = v_k \text{ for } k = 2, \ldots, n.$$

Hence, $T + S = I$ is invertible. Therefore, the set of non-invertible linear maps from V to itself is not a subspace of $\mathcal{L}(V)$. ∎

Exercise 3D5

Suppose V is finite-dimensional, U is a subspace of V, and $S \in \mathcal{L}(U, V)$. Prove that there exists an invertible linear map T from V to itself such that $Tu = Su$ for every $u \in U$ if and only if S is injective.

Solution. Suppose there exists an invertible linear map T from V to itself such that $Tu = Su$ for every $u \in U$. If $Su = 0$, then $Tu = 0$, and since T is invertible, $u = 0$. Hence, S is injective.

Conversely, suppose S is injective. Let $u_1, \ldots, u_m$ be a basis of U. Since S is injective, then $Su_1, \ldots, Su_m$ is linearly independent. Extend $u_1, \ldots, u_m$ to a basis $u_1, \ldots, u_n$ of V. Extend $Su_1, \ldots, Su_m$ to a basis $v_1, \ldots, v_n$ of V. Define $T \in \mathcal{L}(V)$ such that

$$Tu_k = Su_k \text{ for } k = 1, \ldots, m, \quad Tu_k = v_k \text{ for } k = m + 1, \ldots, n.$$

Then T is invertible, since $Tu_1, \ldots, Tu_n$ is a basis of V. Thus, there exists an invertible linear map T from V to itself such that $Tu = Su$ for every $u \in U$. ∎

Exercise 3D6

Suppose that W is finite-dimensional and $S, T \in \mathcal{L}(V, W)$. Prove that $\operatorname{null} S = \operatorname{null} T$ if and only if there exists an invertible $E \in \mathcal{L}(W)$ such that $S = ET$.

Solution. Suppose $\operatorname{null} S = \operatorname{null} T$. Then $\dim \operatorname{range} S = \dim \operatorname{range} T$. Define a linear map E from $\operatorname{range} T$ to $\operatorname{range} S$ by setting $E(Tv) = Sv$ for all $v \in V$. This is well-defined because if $Tv = Tv'$ for some $v, v' \in V$, then $T(v - v') = 0$, so $v - v' \in \operatorname{null} T = \operatorname{null} S$, and thus $S(v - v') = 0$, which implies $Sv = Sv'$. Extend E to an invertible linear map on W. Then $S = ET$.

Conversely, suppose there exists an invertible $E \in \mathcal{L}(W)$ such that $S = ET$. If $v \in \operatorname{null} S$, then $Sv = 0$, and since $S = ET$, we have $ETv = 0$. Since E is invertible, then $Tv = 0$, and thus $v \in \operatorname{null} T$. Hence, $\operatorname{null} S \subseteq \operatorname{null} T$. Similarly, if $v \in \operatorname{null} T$, then $Tv = 0$, and since $S = ET$, we have $Sv = ETv = E0 = 0$. Thus, $v \in \operatorname{null} S$, and $\operatorname{null} T \subseteq \operatorname{null} S$. Therefore, $\operatorname{null} S = \operatorname{null} T$. ■

Exercise 3D7

Suppose that V is finite-dimensional and $S, T \in \mathcal{L}(V, W)$. Prove that $\operatorname{range} S = \operatorname{range} T$ if and only if there exists an invertible $E \in \mathcal{L}(V)$ such that $S = TE$.

Solution. Suppose $\operatorname{range} S = \operatorname{range} T$. Then $\dim \operatorname{null} S = \dim \operatorname{null} T$. Let $u_1, \dots, u_k$ be a basis of $\operatorname{null} S$ and $v_1, \dots, v_k$ be a basis of $\operatorname{null} T$. Extend both bases to a basis of V, say $u_1 \dots, u_n$ and $v_1, \dots, v_n$. Observe that for $i > k$, we have that $Su_i \in \operatorname{range} S = \operatorname{range} T$. Then there must exist $y_i \in V$ such that $Ty_i = Su_i$. Define a linear map $E \in \mathcal{L}(V)$ by setting $Eu_i = v_i$ for $i \leq k$ and $Eu_i = y_i$ for $i > k$. To see that $TE = S$, we have for $i \leq k$ that $TEu_i = Tv_i = 0 = Su_i$. Moreover, for $i > k$, we have $TEu_i = Ty_i = Su_i$. To see that E is invertible, suppose we have scalars $\alpha_1, \dots, \alpha_n$ such that

$$\alpha_1 Eu_1 + \alpha_2 Eu_2 + \cdots + \alpha_n Eu_n = 0.$$

Applying T, we get

$$\alpha_{k+1} Su_{k+1} + \cdots + \alpha_n Su_n = 0,$$

where we used the fact that $TE = S$ and $Su_1 = \cdots = Su_k = 0$ because $u_1, \dots, u_k \in \operatorname{null} S$. Since $Su_{k+1}, \dots, Su_n$ are linearly independent (they form a basis of $\operatorname{range} S$), we must have $\alpha_{k+1} = \cdots = \alpha_n = 0$. Then from the previous equation, $\alpha_1 v_1 + \cdots + \alpha_k v_k = 0$, and since $v_1, \dots, v_k$ are linearly independent, we must have $\alpha_1 = \cdots = \alpha_k = 0$. Hence, all the scalars are zero, proving that E is invertible.

Conversely, suppose there exists an invertible $E \in \mathcal{L}(V)$ such that $S = TE$. If $w \in \operatorname{range} S$, then there exists some $v \in V$ such that $Sv = w$. Since $S = TE$, we have $TEv = w$, and thus $w \in \operatorname{range} T$. Hence, $\operatorname{range} S \subseteq \operatorname{range} T$. Similarly, if $w \in \operatorname{range} T$, then there exists some $v \in V$ such that $Tv = w$. Since E is invertible, there exists some $u \in V$ such that $Eu = v$. Then we have

$$Su = TEu = Tv = w,$$

and thus $w \in \operatorname{range} S$. Hence, $\operatorname{range} T \subseteq \operatorname{range} S$. Therefore, $\operatorname{range} S = \operatorname{range} T$. ■

Exercise 3D8

Suppose V and W are finite-dimensional and $S, T \in \mathcal{L}(V, W)$. Prove that there exist invertible $E_1 \in \mathcal{L}(V)$ and $E_2 \in \mathcal{L}(W)$ such that $S = E_2 T E_1$ if and only if $\dim \operatorname{null} S = \dim \operatorname{null} T$.

Solution. Suppose there exist invertible $E_1 \in \mathcal{L}(V)$ and invertible $E_2 \in \mathcal{L}(W)$ such that $S = E_2TE_1$. We claim that E_1 is an invertible map from $\operatorname{null} S$ to $\operatorname{null} T$. To see this, suppose $v, w \in \operatorname{null} S$ and $E_1v = E_1w$. Then $v = w$ since E_1 is invertible, proving injectivity. To see surjectivity, let $u \in \operatorname{null} T$. Since E_1 is invertible, there exists some $v \in V$ such that $E_1v = u$. Then

$$Sv = E_2TE_1v = E_2Tu = E_20 = 0,$$

and thus $v \in \operatorname{null} S$. Hence, E_1 is an invertible map from $\operatorname{null} S$ to $\operatorname{null} T$, and thus $\dim \operatorname{null} S = \dim \operatorname{null} T$.

Conversely, suppose $\dim \operatorname{null} S = \dim \operatorname{null} T$. Then we may construct an invertible linear map $E_1 \in \mathcal{L}(V)$ that maps $\operatorname{null} S$ onto $\operatorname{null} T$. We now claim that $\operatorname{null} TE_1 = \operatorname{null} S$. To see this, let $v \in \operatorname{null} TE_1$. This is only true when $TE_1v = 0$, or $E_1v \in \operatorname{null} T$. By invertibility of E_1, we have $v \in \operatorname{null} S$. By Exercise 3D6 with $\operatorname{null} S = \operatorname{null} TE_1$, there exists an invertible $E_2 \in \mathcal{L}(W)$ such that $S = E_2TE_1$. ■

Exercise 3D9

Suppose V is finite-dimensional and $T : V \to W$ is a surjective linear map of V onto W. Prove that there is a subspace U of V such that $T|_U$ is an isomorphism of U onto W.
Axler Note. *Here $T|_U$ means the function T restricted to U. Thus $T|_U$ is the function whose domain is U, with $T|_U(u) = Tu$ for every $u \in U$.*

Solution. Since T is surjective, then $\operatorname{range} T = W$. By Exercise 3B11, there exists a subspace U of V such that $V = \operatorname{null} T \oplus U$. We show that $T|_U$ is an isomorphism of U onto W.

Suppose $u \in U$ such that $T|_Uu = 0$. Then $Tu = 0$, and thus $u \in \operatorname{null} T \cap U$. Since $V = \operatorname{null} T \oplus U$, we have $\operatorname{null} T \cap U = \{0\}$, and thus $u = 0$. This shows that $T|_U$ is injective.

To see surjectivity, let $w \in W$. Since T is surjective, there exists $v \in V$ such that $Tv = w$. Write $v = n + u$ with $n \in \operatorname{null} T$ and $u \in U$. Then $Tu = T(v - n) = Tv - Tn = w - 0 = w$. Hence, $T|_U$ is surjective onto W, and therefore an isomorphism of U onto W. ■

Exercise 3D10

Suppose V and W are finite-dimensional and U is a subspace of V. Let

$$\mathcal{E} = \{T \in \mathcal{L}(V, W) \mid U \subseteq \operatorname{null} T\}.$$

(a) Show that $\mathcal{E}$ is a subspace of $\mathcal{L}(V, W)$.

(b) Find a formula for $\dim \mathcal{E}$ in terms of $\dim V$, $\dim W$, and $\dim U$.

Axler Hint. *Define $\Phi : \mathcal{L}(V, W) \to \mathcal{L}(U, W)$ by $\Phi(T) = T|_U$. What is $\operatorname{null} \Phi$? What is $\operatorname{range} \Phi$?*

Solution.

(a) Consider the zero map $0 \in \mathcal{L}(V, W)$. For this map, $U \subseteq \operatorname{null} 0$ since $\operatorname{null} 0 = V$. Hence, $0 \in \mathcal{E}$, showing that $\mathcal{E}$ is non-empty. To see that $\mathcal{E}$ is closed under addition and scalar multiplication, let $S, T \in \mathcal{E}$ and $\alpha \in \mathbb{F}$. For any $u \in U$, we have $(S + T)u = Su + Tu = 0 + 0 = 0$, so $S + T \in \mathcal{E}$. Similarly, $(\alpha T)u = \alpha(Tu) = \alpha \cdot 0 = 0$, so $\alpha T \in \mathcal{E}$. Therefore, $\mathcal{E}$ is a subspace of $\mathcal{L}(V, W)$.

(b) Consider the linear map $\Phi : \mathcal{L}(V, W) \to \mathcal{L}(U, W)$ defined by $\Phi(T) = T|_U$. Let $T \in \mathcal{L}(V, W)$ such that $\Phi(T) = T|_U = 0$. Then $T|_U = 0$ implies that for every $u \in U$, we have $Tu = 0$, i.e., $U \subseteq \operatorname{null} T$. Hence, $T \in \mathcal{E}$, showing that $\operatorname{null} \Phi \subseteq \mathcal{E}$. Conversely, if $T \in \mathcal{E}$, then $U \subseteq \operatorname{null} T$, which means that $T|_U = 0$, i.e., $T \in \operatorname{null} \Phi$. Therefore, $\operatorname{null} \Phi = \mathcal{E}$, and $\dim \mathcal{E} = \dim \operatorname{null} \Phi$.

We now show that Φ is surjective. Let $S \in \mathcal{L}(U, W)$, and let $u_1, \ldots, u_m$ be a basis for U. Extend this to a basis $u_1, \ldots, u_m, v_1, \ldots, v_n$ for V. Define a linear map $T \in \mathcal{L}(V, W)$ by setting $Tu_i = Su_i$ for $i = 1, \ldots, m$ and $Tv_j = 0$ for $j = 1, \ldots, n$. Then $T|_U = S$, so $\Phi(T) = S$. Hence, Φ is surjective, and $\operatorname{range} \Phi = \mathcal{L}(U, W)$. Therefore, $\dim \operatorname{range} \Phi = \dim \mathcal{L}(U, W) = (\dim U)(\dim W)$. By the rank-nullity theorem,

$$\dim \mathcal{L}(V, W) = \dim \operatorname{null} \Phi + \dim \operatorname{range} \Phi = \dim \mathcal{E} + (\dim U)(\dim W),$$

Observing that $\dim \mathcal{L}(V, W) = (\dim V)(\dim W)$, we get

$$\dim \mathcal{E} = (\dim V)(\dim W) - (\dim U)(\dim W) = (\dim V - \dim U)(\dim W).$$

■

Exercise 3D11

Suppose V is finite-dimensional and $S, T \in \mathcal{L}(V)$. Prove that

$$ST \text{ is invertible} \iff S \text{ and } T \text{ are invertible.}$$

Solution. Suppose ST is invertible. Then it is injective. By Exercise 3B19, there exists $U \in \mathcal{L}(V)$ such that $U(ST)$ is the identity operator. Since this is equal to $(US)T$, this implies that T is injective. Since T is injective on $\mathcal{L}(V)$, it is also surjective, hence invertible.

Note that ST is also surjective. By Exercise 3B20, there exists $W \in \mathcal{L}(V)$ such that STW is the identity operator. Then $S(TW)$ implies that S is surjective. Since it is surjective on V, then S is also injective, hence invertible.

Conversely, suppose S and T are invertible. This follows from Exercise 3D2. ■

Exercise 3D12

Suppose V is finite-dimensional and $S, T, U \in \mathcal{L}(V)$ and $STU = I$. Show that T is invertible and that $T^{-1} = US$.

Solution. Since $STU = I$, then we have that $S(TU) = I$. By Exercise 3B20, then S is surjective, hence injective. Then S is invertible with $S^{-1} = TU$. Similarly, $(ST)U = I$. By Exercise 3B19, then ST is injective, hence surjective. Then ST is invertible with $(ST)^{-1} = U$. Then

$$T(US) = (TU)S = S^{-1}S = I, \quad (US)T = U(ST) = (ST)^{-1}(ST) = I.$$

Hence, T is invertible with $T^{-1} = US$. ■

Exercise 3D13

Show that the result in Exercise 3D12 can fail without the hypothesis that V is finite-dimensional.

Solution. If V were not finite-dimensional, let $v_1, v_2, \ldots$ be an infinite basis for V. Let $S, T, U \in \mathcal{L}(V)$ be defined as follows: let $S = I$,

$$Tv_1 = 0, \quad Tv_k = v_{k-1} \text{ for } k = 2, 3, \ldots$$

and

$$Uv_k = v_{k+1} \text{ for } k = 1, 2, \ldots$$

It is clear that $TUv_k = v_k$ for all $k = 1, 2, \ldots$, so $STU = I$. However, T is not injective since $\operatorname{null} T = \operatorname{span} v_1$, and thus T is not invertible. ■

Exercise 3D14

Prove or give a counterexample: If V is a finite-dimensional vector space and $R, S, T \in \mathcal{L}(V)$ are such that RST is surjective, then S is injective.

Solution. Since RST is surjective, and V is finite dimensional, it follows that RST is injective, hence invertible. By Exercise 3D11, then $RST = (RS)T$ is invertible if and only if RS and T are invertible. In particular, RS is invertible, and applying Exercise 3D11 again, we have that R and S are invertible. Hence, S is injective. ■

Exercise 3D15

Suppose $T \in \mathcal{L}(V)$ and $v_1, \ldots, v_m$ is a list in V such that $Tv_1, \ldots, Tv_m$ spans V. Prove that $v_1, \ldots, v_m$ spans V.

Solution. Since $Tv_1, \ldots, Tv_m$ spans V, and V is finite-dimensional, then T is invertible. Then for every $v \in V$, there exist scalars $\alpha_1, \ldots, \alpha_m$ such that

$$v = \alpha_1 Tv_1 + \cdots + \alpha_m Tv_m.$$

Applying T^{-1} to both sides, we have

$$T^{-1}v = \alpha_1 v_1 + \cdots + \alpha_m v_m.$$

Since v was arbitrary, then $v_1, \ldots, v_m$ spans V. ■

Exercise 3D16

Prove that every linear map from $\mathbb{F}^{n,1}$ to $\mathbb{F}^{m,1}$ is given by a matrix multiplication. In other words, prove that if $T \in \mathcal{L}(\mathbb{F}^{n,1}, \mathbb{F}^{m,1})$, then there exists an m-by-n matrix A such that $Tx = Ax$ for every $x \in \mathbb{F}^{n,1}$.

Solution. Let $e_1, \ldots, e_n$ be the standard basis of $\mathbb{F}^{n,1}$. Define the k-th column of A to be $Te_k \in \mathbb{F}^{m,1}$ for $k = 1, \ldots, n$. Then for every $x \in \mathbb{F}^{n,1}$, there exist scalars $\alpha_1, \ldots, \alpha_n$ such that

$$x = \alpha_1 e_1 + \cdots + \alpha_n e_n.$$

Applying T to both sides, we have

$$Tx = \alpha_1 Te_1 + \cdots + \alpha_n Te_n = Ax,$$

where the last equality follows from the definition of matrix multiplication. Hence, $Tx = Ax$ for every $x \in \mathbb{F}^{n,1}$. ■

Exercise 3D17

Suppose V is finite-dimensional and $S \in \mathcal{L}(V)$. Define $\mathcal{A} \in \mathcal{L}(\mathcal{L}(V))$ by

$$\mathcal{A}(T) = ST$$

for $T \in \mathcal{L}(V)$.

(a) Show that $\dim \operatorname{null} \mathcal{A} = (\dim V)(\dim \operatorname{null} S)$.

(b) Show that $\dim \operatorname{range} \mathcal{A} = (\dim V)(\dim \operatorname{range} S)$.

Solution.

(a) Consider $T \in \operatorname{null} \mathcal{A}$. Then $\mathcal{A}(T) = ST = 0$, so T is a linear map from V to $\operatorname{null} S$. Conversely, if T is a linear map from V to $\operatorname{null} S$, then $ST = 0$, so $T \in \operatorname{null} \mathcal{A}$. Hence, $\operatorname{null} \mathcal{A}$ is the set of linear maps from V to $\operatorname{null} S$, and thus $\dim \operatorname{null} \mathcal{A} = (\dim V)(\dim \operatorname{null} S)$.

(b) Consider $T \in \operatorname{range} \mathcal{A}$. Then there exists some $U \in \mathcal{L}(V)$ such that $\mathcal{A}(U) = ST$. Since ST is a linear map from V to $\operatorname{range} S$, then $\operatorname{range} \mathcal{A}$ is the set of linear maps from V to $\operatorname{range} S$, and thus $\dim \operatorname{range} \mathcal{A} = (\dim V)(\dim \operatorname{range} S)$. ■

Exercise 3D18

Show that V and $\mathcal{L}(\mathbb{F}, V)$ are isomorphic vector spaces.

Solution. Let $T : \mathcal{L}(\mathbb{F}, V) \to V$ be defined by $T(\alpha) = \alpha(1)$ for $\alpha \in \mathcal{L}(\mathbb{F}, V)$. We first show that T is a linear map. Let $\alpha, \beta \in \mathcal{L}(\mathbb{F}, V)$ and $\lambda \in \mathbb{F}$. Then

$$T(\alpha + \beta) = (\alpha + \beta)(1) = \alpha(1) + \beta(1) = T\alpha + T\beta,$$

and

$$T(\lambda\alpha) = (\lambda\alpha)(1) = \lambda\alpha(1) = \lambda T\alpha.$$

We now show that T is an isomorphism. To see injectivity, let $\alpha, \beta \in \mathcal{L}(\mathbb{F}, V)$ such that $T\alpha = T\beta$. Then $\alpha(1) = \beta(1)$. For any $x \in \mathbb{F}$, we have $\alpha(x) = x\alpha(1) = x\beta(1) = \beta(x)$, so $\alpha = \beta$. Hence, T is injective.

To see surjectivity, let $v \in V$. Define $\alpha \in \mathcal{L}(\mathbb{F}, V)$ by $\alpha(x) = xv$ for $x \in \mathbb{F}$. Then $T\alpha = \alpha(1) = v$, so T is surjective. Therefore, T is an isomorphism, and thus V and $\mathcal{L}(\mathbb{F}, V)$ are isomorphic vector spaces. ■

Exercise 3D19

Suppose V is finite-dimensional and $T \in \mathcal{L}(V)$. Prove that T has the same matrix with respect to every basis of V if and only if T is a scalar multiple of the identity operator.
Manual Remark. *This exercise has a beautiful alternate solution as follows: let V have a basis $v_1, \ldots, v_n$. Consider the basis $w_1, \ldots, w_n$, where $w_i = v_j$ and $w_j = v_i$ for some $i \neq j$, and $w_k = v_k$ for $k \neq i, j$. One may perform similar calculations to deduce that the diagonal entries are equal to some λ, while every off-diagonal entry must be equal to some γ. The last step is to consider the basis $-v_1, v_2, \ldots, v_n$ to deduce that $\gamma = 0$, so that T is a scalar multiple of the identity operator.*

Solution. Suppose T has the same matrix with respect to every basis of V. Then for any set of bases for V, its matrix must be the same. In particular, for any bases $v_1, \ldots, v_n$ and $u_1, \ldots, u_n$ of V, we have

$$\mathcal{M}(T, (v_1, \ldots, v_n)) = \mathcal{M}(T, (u_1, \ldots, u_n)).$$

Hence, we can declare that their matrix is just some A. We now consider the basis $v_1 + v_2, v_2, \ldots, v_n$, which yields the same matrix A. Let $w_1 = v_1 + v_2$ and $w_i = v_i$ for $i \geq 2$. In the new basis, we have the expression

$$Tw_1 = A_{1,1}w_1 + A_{2,1}w_2 + \cdots + A_{n,1}w_n = A_{1,1}v_1 + (A_{1,1} + A_{2,1})v_2 + A_{3,1}v_3 + \cdots + A_{n,1}v_n.$$

This must equal the expression for $Tw_1 = Tv_1 + Tv_2$ in the original basis:

$$Tv_1 + Tv_2 = (A_{1,1} + A_{1,2})v_1 + (A_{2,1} + A_{2,2})v_2 + \cdots + (A_{n,1} + A_{n,2})v_n.$$

Comparing coefficients of $v_1, \ldots, v_n$ in the two expressions, we get

$$\begin{aligned} A_{1,1} &= A_{1,1} + A_{1,2}, \\ A_{1,1} + A_{2,1} &= A_{2,1} + A_{2,2}, \\ A_{j,1} &= A_{j,1} + A_{j,2} \text{ for } j \geq 3. \end{aligned}$$

From this, we obtain that $A_{1,2} = 0, A_{2,2} = A_{1,1}$, and $A_{j,2} = 0$ for $j \geq 3$, i.e., the diagonal elements $A_{1,1}$ and $A_{2,2}$ are equal, while off-diagonal entries in column 2 are 0. Now consider the basis $v_1, v_1 + v_2, \ldots, v_n$. Let $u_1 = v_1, u_2 = v_1 + v_2$, and $u_i = v_i$ for $i \geq 3$. In the new basis, we have

$$\begin{aligned} Tu_2 &= A_{1,2}u_1 + \cdots + A_{n,2}u_n \\ &= A_{1,2}v_1 + A_{2,2}(v_1 + v_2) + A_{3,2}v_3 + \cdots + A_{n,2}v_n \\ &= (A_{1,2} + A_{2,2})v_1 + A_{2,2}v_2 + A_{3,2}v_3 + \cdots + A_{n,2}v_n. \\ &= A_{1,1}v_1 + A_{2,2}v_2. \end{aligned}$$

This must equal the expression for $Tu_2 = Tv_1 + Tv_2$ in the original basis, whose expression is listed above. Comparing coefficients of v_1 and v_2, we get

$$\begin{aligned} A_{1,1} &= A_{1,1} + A_{1,2}, \\ A_{2,2} &= A_{2,1} + A_{2,2}, \\ A_{j,2} &= A_{j,1} + A_{j,2} \quad \text{for } j \geq 3. \end{aligned}$$

From this, we obtain similar results about diagonal entries, but the off-diagonal entries in column 1 must also be 0. We may repeat the first process for $v_1 + v_k$ for $k \geq 3$ to get that entries off the diagonal in column k must also be 0. Continuing this process, we conclude all off-diagonal entries are 0, and all diagonal entries are equal, i.e., A is a scalar multiple of the identity matrix.

Conversely, suppose T is a scalar multiple of the identity operator. Then there exists some $\lambda \in \mathbb{F}$ such that $T = \lambda I$. For any basis of V, the matrix representation of T with respect to that basis is the diagonal matrix with λ on the diagonal. Hence, T has the same matrix with respect to every basis of V. ■

Exercise 3D20

Suppose $q \in \mathcal{P}(\mathbb{R})$. Prove that there exists a polynomial $p \in \mathcal{P}(\mathbb{R})$ such that

$$q(x) = (x^2 + x)p''(x) + 2xp'(x) + p(3)$$

for all $x \in \mathbb{R}$.

Solution. Let $T : \mathcal{P}_n(\mathbb{R}) \to \mathcal{P}_n(\mathbb{R})$ be defined by

$$Tp = (x^2 + x)p'' + 2xp' + p(3)$$

for $p \in \mathcal{P}_n(\mathbb{R})$. We first show that T is a linear map. Let $p, q \in \mathcal{P}_n(\mathbb{R})$ and $\alpha \in \mathbb{R}$. Then

$$T(p + q) = (x^2 + x)(p + q)'' + 2x(p + q)' + (p + q)(3) = Tp + Tq,$$

and

$$T(\alpha p) = (x^2 + x)(\alpha p)'' + 2x(\alpha p)' + (\alpha p)(3) = \alpha Tp.$$

Then $T \in \mathcal{L}(\mathcal{P}_n(\mathbb{R}))$. We now show that T is invertible. To see injectivity, let $p \in \mathcal{P}_n(\mathbb{R})$ such that $Tp = 0$. Then

$$(x^2 + x)p'' + 2xp' + p(3) = 0.$$

Then

$$(x^2 + x)p'' + 2xp' = -p(3).$$

We analyze the degree of p as follows:

- If $\deg p \geq 2$, consider $(x^2 + x)p''$. In particular, $(a_m x^m)'' = a_m m(m-1)x^{m-2}$, hence the coefficient of x^m is $a_m m(m-1)$. Moreover, $2xp'$ has coefficient $2ma_m$. Since $\deg p \geq 2$, then $m \geq 2$, so the total coefficient of x^m is $a_m m(m-1) + 2ma_m = a_m m(m-1+2) = a_m m(m+1)$. Since this is nonzero for $a_m \neq 0$, we get a contradiction with the right-hand side being a constant.

- If $\deg p = 1$, then $p(x) = ax + b$ for some $a, b \in \mathbb{R}$. Then $p''(x) = 0$ and $p'(x) = a$, so the left-hand side is $2ax$. Since the right-hand side is a constant, this is a contradiction.

- If $\deg p = 0$, then $p(x) = c$ for some $c \in \mathbb{R}$. Then $p''(x) = 0$ and $p'(x) = 0$, so the left-hand side is 0, and the right-hand side is $p(3) = c$. Hence, $c = 0$, and thus $p = 0$. Therefore, T is injective.

Since T is injective and $\mathcal{P}_n(\mathbb{R})$ is finite-dimensional, then T is surjective, hence invertible. Then there does exist some $p \in \mathcal{P}_n(\mathbb{R})$ such that $Tp = q$. ■

Exercise 3D21

Suppose n is a positive integer and $A_{j,k} \in \mathbb{F}$ for all $j,k = 1,\dots,n$. Prove that the following are equivalent (note that in both parts below, the number of equations equals the number of variables).

(a) The trivial solution $x_1 = \cdots = x_n = 0$ is the only solution to the homogeneous system of equations

$$\sum_{k=1}^{n} A_{1,k}x_k = 0$$
$$\vdots$$
$$\sum_{k=1}^{n} A_{n,k}x_k = 0$$

(b) For every $c_1,\dots,c_n \in \mathbb{F}$, there exists a solution to the system of equations

$$\sum_{k=1}^{n} A_{1,k}x_k = c_1$$
$$\vdots$$
$$\sum_{k=1}^{n} A_{n,k}x_k = c_n$$

Solution.

(a) Suppose the trivial solution $0 \in \mathbb{F}^{n,1}$ is the only solution to the homogeneous system of equations. Then $\operatorname{null} T = \{0\}$, where $T : \mathbb{F}^{n,1} \to \mathbb{F}^{n,1}$ is defined by $Tx = Ax$ for $x \in \mathbb{F}^{n,1}$. Since $\operatorname{null} T = \{0\}$, then T is injective. Since $\mathbb{F}^{n,1}$ is finite-dimensional, then T is surjective, hence invertible. Then for every $c_1,\dots,c_n \in \mathbb{F}$, there exists some $x \in \mathbb{F}^{n,1}$ such that $Tx = c$, where c is the column vector with entries $c_1,\dots,c_n$. Hence, there exists a solution to the system of equations.

(b) Suppose for every $c_1,\dots,c_n \in \mathbb{F}$, there exists a solution to the system of equations. Then $T : \mathbb{F}^{n,1} \to \mathbb{F}^{n,1}$ defined by $Tx = Ax$ for $x \in \mathbb{F}^{n,1}$ is surjective. Since $\mathbb{F}^{n,1}$ is finite-dimensional, then T is injective. Hence, the trivial solution is the only solution to the homogeneous system of equations. ∎

Exercise 3D22

Suppose $T \in \mathcal{L}(V)$ and $v_1,\dots,v_n$ is a basis of V. Prove that

$$\mathcal{M}(T,(v_1,\dots,v_n)) \text{ is invertible} \iff T \text{ is invertible.}$$

Solution. Suppose $\mathcal{M}(T,(v_1,\dots,v_n))$ is invertible. Then there exists some $A \in \mathbb{F}^{n,n}$ such that $A\mathcal{M}(T,(v_1,\dots,v_n)) = I$. Let U be the linear map from V to V whose matrix with respect to the basis $v_1, \dots, v_n$ is A. Then

$$\mathcal{M}(UT, (v_1, \dots, v_n)) = \mathcal{M}(U, (v_1, \dots, v_n))\mathcal{M}(T, (v_1, \dots, v_n)) = A\mathcal{M}(T, (v_1, \dots, v_n)) = I.$$

Then UT is the identity operator, and thus T is injective. Since V is finite-dimensional, then T is surjective, hence invertible.

Suppose T is invertible. Then T^{-1} exists, and we have

$$\mathcal{M}(T^{-1}T, (v_1, \dots, v_n)) = \mathcal{M}(T^{-1}, (v_1, \dots, v_n))\mathcal{M}(T, (v_1, \dots, v_n)) = I.$$

Hence, $\mathcal{M}(T, (v_1, \dots, v_n))$ is invertible. ■

Exercise 3D23

Suppose that $u_1, \dots, u_n$ and $v_1, \dots, v_n$ are bases of V. Let $T \in \mathcal{L}(V)$ be such that $Tv_k = u_k$ for each $k = 1, \dots, n$. Prove that

$$\mathcal{M}(T, (v_1, \dots, v_n)) = \mathcal{M}(I, (u_1, \dots, u_n), (v_1, \dots, v_n)).$$

Solution. By definition, column k of $\mathcal{M}(T, (v_1, \dots, v_n))$ is $Tv_k = u_k$ with respect to the basis $v_1, \dots, v_n$. Similarly, column k of $\mathcal{M}(I, (u_1, \dots, u_n), (v_1, \dots, v_n))$ is also u_k with respect to the basis $v_1, \dots, v_n$. ■

Exercise 3D24

Suppose A and B are square matrices of the same size and $AB = I$. Prove that $BA = I$.

Solution. Let $T : \mathbb{F}^{n,1} \to \mathbb{F}^{n,1}$ be defined by $Tx = Ax$ for $x \in \mathbb{F}^{n,1}$. Since $AB = I$, then T is surjective. Since $\mathbb{F}^{n,1}$ is finite-dimensional, surjectivity implies injectivity. Hence, T is invertible, and $T(Bx) = x = T(T^{-1}x)$. Then $Bx = T^{-1}x$ for every $x \in \mathbb{F}^{n,1}$. Therefore,

$$BAx = B(Tx) = T^{-1}(Tx) = x$$

for every $x \in \mathbb{F}^{n,1}$. Hence, $BA = I$. ■